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PALAEOMAGNETISM OF THE SVECOFENNIAN LOFTAHAMMAR GABBRO AND SOME JOTNIAN DOLERITES IN THE SWEDISH PART OF THE BALTIC SHIELD

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The Svecofennian Loftahammar gabbro (Rb–Sr isochron age $1{,}694 \cdot 10^6$ year) and Jotnian dolerites (K–Ar isochron age $1{,}245 \cdot 10^6$ year) in Sweden were found to have palaeomagnetic pole positions in close agreement with poles from other Svecofennian rocks in Sweden and Jotnian dolerites in Finland, respectively. These results support Neuvonen's apparent polar wandering path during the period from $1{,}900 \cdot 10^6$ to $1{,}300 \cdot 10^6$ year.

A reconstruction of the position of the Baltic Shield during the $(1{,}900–1{,}200) \cdot 10^6$-year interval, based on available pole positions, indicates that the Baltic shield was close to Greenland and the North American Shield and reveals a contiguity of North Atlantic geologic-geochronological zones until $1{,}200 \cdot 10^6$ year ago.

1. Introduction

The Baltic Shield can be subdivided in four geologic-geochronological zones, indicated in the upper inset of Fig. 1. The corresponding periods of orogeny and metamorphism are compiled in Table I. This paper presents results from a palaeomagnetic study of samples collected from rocks belonging to the Svecofennian zone, also called Svecokarelian zone, and from post-Svecofennian, Jotnian dolerites.

Mulder (1971) has investigated Svecofennian and Jotnian rocks from the Dalarna district of Sweden, whereas Neuvonen has studied the palaeomagnetism of Jotnian rocks from SW Finland (Neuvonen, 1965, 1966, 1970, 1973; Neuvonen and Grundström, 1969). Neuvonen (1970) has suggested an apparent polar wandering path for the Baltic Shield (Fig. 2), which indicates a drift of the Baltic Shield towards the palaeo-equator. Donaldson et al. (1973) used this pole path to reconstruct the position of the Baltic Shield relative to the Canadian Shield, showing a co-linearity of the Grenville front in Canada and the Gothide tectonic boundary in Sweden. In this reconstruction the distance between the two shields is approximately the same as at present.

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2. Palaeomagnetic investigations

This study was concentrated on a Svecofennian gabbro massif in the Västervik area, and on Jotnian dolerites in the coastal district of the Bothnian Sea. Radiometric age determinations are available from both units. In addition, samples were collected from miscellaneous Svecofennian amphibolites, from Svecofennian granitic gneiss in the Västervik area and sub-Jotnian gabbro-anorthosite at Nordingrå. Table II gives a complete list of sampling locations.

Five to nine core samples, 2.5 cm in diameter, were drilled at each site and cut into specimens of 2.2 cm in height. A suncompass was used for the orientation.

The magnetic measurements were carried out with an astatic magnetometer or, in case of low intensities of the natural remanent magnetization (NRM)

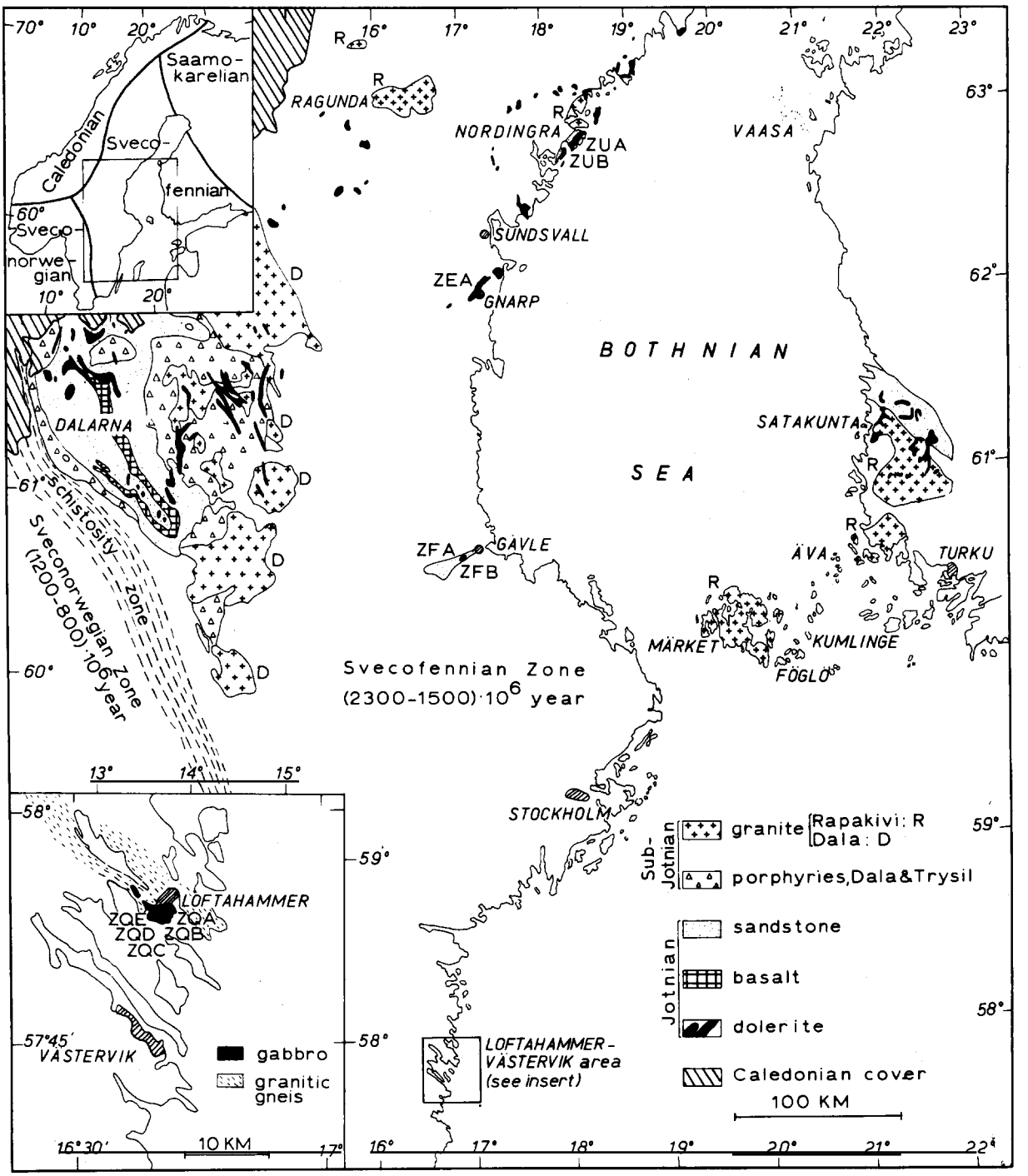


Fig. 1. Geological outlines and sampling locations.

TABLE I

Time-stratigraphic scheme

Age (10^6 year)	
370– 600	Caledonian period of orogeny and metamorphism
800–1,200	Sveconorwegian period of orogeny and metamorphism (Gerling et al., 1968)
740–1,350	K–Ar ages of Jotnian dolerites (Lundqvist, 1968)
850–1,310	K–Ar ages of Jotnian sediments (Lundqvist, 1968)
1,245	K–Ar isochron age Jotnian Nordingrå dolerite (Welin and Lundqvist, 1975)
1,350	sub-Jotnian granite intrusion at Ragunda Welin and Lundqvist, 1975)
1.415	sub-Jotnian granite intrusion at Nordingrå (Welin and Lundqvist, 1975)
1,465	K–Ar ages Västervik complex (Priem et al., 1969)
1,669	sub-Jotnian granites and volcanics in the Dalarna and Trysil districts of Sweden and Norway (Priem et al., 1970)
1,694	Rb–Sr isochron age Västervik complex (Priem and Bakker, 1973)
1,825	granite intrusion in the Västervik complex (Priem and Bakker, 1973)
1,900	gabbro-diorite intrusions Sweden and Finland (Neuvonen, 1974)
1,500–2,100	Svecofennian period of orogeny and metamorphism (Gerling et al., 1968)
1,900–2,900	Saamokarelian period of orogeny and metamorphism (Gerling et al., 1968)
> 2,900	ratiometric ages of Early Precambrian rocks

(10^{-5}–10^{-6} G), with a Czech (JR-3) spinner magnetometer (Jelínek, 1966).

The initial NRM of all samples was measured and of one or two test-specimens from each site the NRM was analysed by progressive a.f. demagnetization. Test-specimens from sites in the Loftahammar gabbro and in the Jotnian dolerites (sites indicated in Fig. 1) were found to have stable characteristic remanent magnetizations. Test-specimens from the miscellaneous Svecofennian amphibolites, the Loftahammar granitic gneiss and sub-Jotnian gabbro-anorthosite possessed either a soft magnetization only, or showed an erratic response of the remanent magnetization to progressive a.f. demagnetization. The samples from these rocks were excluded from further study.

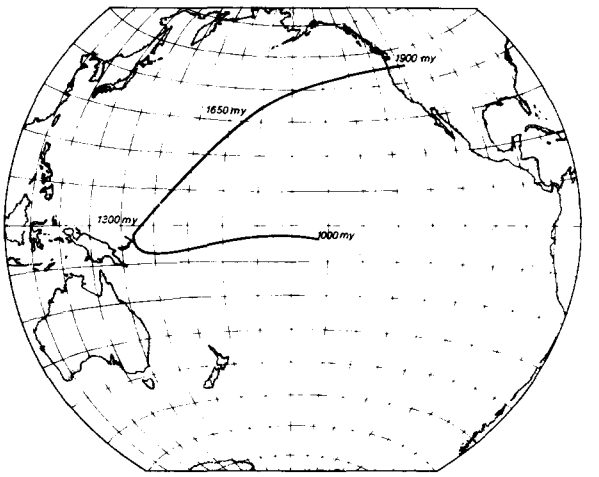


Fig. 2. Apparent polar wandering path for the Baltic Shield after Neuvonen (1970) for the (1,900–1,300) · 10^6-year interval and after Poorter (1975) for the track from 1,300 · 10^6 to 1,000 · 10^6 year.

3. Geological setting

The Västervik area belongs to the Svecofennian zone of orogeny and metamorphism and consists of metasedimentary rocks, surrounded by granitic gneisses. Small dikes and larger massifs of basic rock occur within this area. One of these larger massifs is the Loftahammar gabbro, which is about 4 km in diameter and is considered to be a synorogenic intrusion (Westra et al., 1969). The Västervik area is shown in the lower inset in Fig. 1.

According to Priem and Bakker (1973) the emplacement of the original granitic magma in the Västervik area took place at approximately 1,825 · 10^6 year ago, whereas the gneissification during the main phase of metamorphism is thought to be reflected by a total-rock Rb–Sr isochron age of (1,694 ± 35) · 10^6 year *. According to Priem and Bakker these ages correspond to the Svecofennian period of igneous, tectonic and metamorphic activities. The post-Svecofennian stage of uplift and cooling is possibly reflected by K–Ar and Rb–Sr ages of biotite in the granitic rocks of the Västervik area, which are (1,465 ± 30) · 10^6 and (1,460 ± 50) · 10^6 year, respectively. On the other

* Rb–Sr ages mentioned in this paper are based on ^{87}Rb with a decay constant of 1.39 · 10^{-11}/year.

TABLE II

Location of sampling sites and rocks types

Rock type	Site	No. of samples	Locality		
			lat. (°'N)	long. (°'E)	
Svecofennian Loftahammar hornblende gabbro	*ZQA*	7	57 53	16 41	3 km S of Loftahammar
	ZQB	5	57 54	16 40	1.5 km W of Loftahammar
	ZQC	6	57 54	16 39	2.5 km W of Loftahammar
	ZQD	7	57 54	16 39	2 km W of Loftahammar
	ZQE	6	57 54	16 41	1 km W of Loftahammar
Jotnian dolerites	*ZEA*	6	62 05	17 17	10 km N of Gnarp
	ZFA	6	60 38	16 58	33 km W of Gävle
	ZFB	9	60 38	16 56	35 km W of Gävle
	ZUA	6	62 53	18 24	8 km SE of Nordingrå
	ZUB	6	62 52	18 23	8.5 km SE of Nordingrå
Svecofennian Loftahammar granitic gneiss	*ZOA*	8	57 55	16 35	7 km W of Loftahammar
	ZOB	5	57 54	16 39	3 km W of Loftahammar
Svecofennian amphibolites	*ZEB*	7	60 32	16 44	10 km SSW of Forsbacka
	ZEC	7	61 00	16 43	11 km N of Ockelbo
Sub-Jotnian gabbro-anorthosite	*ZUC*	9	62 55	18 19	1.5 km SE of Nordingrå
	ZUD	7	62 56	18 16	2 km W of Nordingrå
	ZUE	5	62 56	18 15	3 km W of Nordingrå

hand Priem et al. (1969), who reported these ages, suggest that it represents a separate Gothian event. The thermal history of the Västervik area, as deduced from the radiometric evidence, and also applicable to the Loftahammar gabbro, can be described either as a prolonged slow cooling from the thermal maximum attained at $1{,}694 \cdot 10^6$ year, lasting until at least $1{,}465$ 10^6 year, or in terms of a faster cooling and a reheating at approximately $1{,}465 \cdot 10^6$ year ago.

The impact of the Svecofennian deformation of the Loftahammar gabbro was small. Apparently owing to the competent and massive character of the gabbro, old tectonic features in granitic inclusions were preserved during later deformation phases, and linear or planar fabrics in the Loftahammar gabbro are almost absent.

The mineralogical composition of the Loftahammar gabbro is characterized by the paragenesis olivine, ortho- and clinopyroxene, green hornblende and labradorite. In the outer zone of the massif biotite and sulphide minerals become more important. The dominant opaque mineral is magnetite, which may contain lamellae or irregular intergrowths of ilmenite. The TiO_2 content of the magnetite, as determined by microprobe analysis, was generally found to be less than 0.1 wt.%.

The anorogenic stage that followed the Svecofennian period, is characterized by magmatic activity giving rise to emplacement of granites of the Rapakivi type in Finland and Sweden (Nordingrå and Ragunda granites). In the Dalarna and Trysil districts of Sweden and Norway similar granite intrusions were accompanied by porphyries. These anorogenic, called sub-Jotnian intrusions, have widely different Rb–Sr isochron ages. Priem et al. (1970) calculated an over-all Rb–Sr isochron age for the Dala and Trysil granites and porphyries of $(1{,}669 \pm 42) \cdot 10^6$ year, which is based on additional results from Welin and Lundqvist (1970). The age of the granite at Ragunda proved to be $1{,}350 \cdot 10^6$ year and at Nordingrå $(1{,}415 \pm 30) \cdot 10^6$ year (Welin and Lundqvist, 1975). The sediments that have been deposited on the erosion surfaces of these granitic rocks, the Jotnian sandstones, are therefore younger than $1{,}350 \cdot 10^6$ year. The lopolith-shaped Nordingrå

dolerite, intrusive in these Jotnian sandstones, has a K–Ar isochron age of (1,245 ± 20) · 10^6 year (Welin and Lundqvist, 1975). Priem et al. (1968) found K–Ar ages of Jotnian dolerites in the Dalarna district in the range of 745 · 10^6 year to 1,100 · 10^6 year, and discussed the younger ages in terms of Ar loss during the Sveconorwegian and Palaeozoic orogenies, suggesting the age of the dolerites to be greater than 945 · 10^6 years.

The Jotnian sediments and dolerites are in general tectonically undeformed. Thus, in palaeomagnetic studies of these rocks the directions of the remanent magnetization need no correction for tectonical tilt (Mulder, 1971; Neuvonen, 1973). At Nordingrå the Jotnian sandstone dips 5° towards the southeast. Lundqvist and Samuelsson (1973) investigated the mineralogical and chemical variation within the 200 m thick dolerite sheet which rests conformable on this sandstone. Our samples from the Nordingrå dolerite were collected at sites *ZUA* and *ZUB*, about 3.5 km SE of the area investigated byLundqvist and Samuelsson. These samples are considered to be from the top of the dolerite sheet; this appears from the composition of the olivines (Table III), which matches olivines in the top of the section studied by Lundqvist and Samuelsson (1973). According to the differentiation model developed by these authors for the Nordingrå dolerite, the upper part probably crystallized later than the middle and lower part.

The dolerite at Gnarp is assumed to form an extension of the dolerite at Nordingrå. The olivine composition of this dolerite corresponds to that found in the lower and middle part of the Nordingrå dolerite, and suggests a common differentiation state. The mineral compositions determined in the dolerite samples collected at Gävle are comparable with those found at Gnarp and in the lower and middle part of the dolerite sheet at Nordingrå.

The Fe–Ti-oxide minerals in the samples from the Jotnian dolerites are ilmenite and titaniferous magnetite. The description of the Fe–Ti-oxide minerals in the dolerite at Nordingrå, given by Lundqvist and Samuelsson (1973) is pertinent to our samples collected at sites *ZUA*, *ZUB* and *ZEA* (Nordingrå and Gnarp). According to these authors the titaniferous magnetite shows exsolution in magnetite and ulvöspinel, which are greyish and light brown in colour (Mogensen, 1946). In addition, magnetite and ilmenite intergrowths are described, which are explained by supposing the ulvöspinel being oxidated to ilmenite and magnetite (Buddington and Lindsley, 1964) in particular in the upper part of the dolerite sheet. Identical phenomena were observed in our samples from sites *ZUA* and *ZUB* and have been also described by Neuvonen (1965) for the Jotnian olivine dolerites at Satakunta in Finland. The dolerite at Gnarp (site *ZEA*) shows the same fine-scale (1 μm and less) intergrowth of ulvöspinel and magnetite, the so-called "cloth-texture" (Vincent and Phillips, 1954) as the dolerite at Nordingrå, but the oxidation-type intergrowth of magnetite and ilmenite was not found. Samples from the dolerite at Gävle (sites *ZFA* and *ZFB*) appear to have homogeneous titaniferous magnetite (magnification 1,600X). A submicroscopic exsolution of this magnetite, however, cannot be excluded. Microprobe investigation of the

TABLE III

Compositions of olivine, augite and plagioclase in specimens from Jotnian dolerites (locations are listed in Table II); the compositions were calculated from partial chemical microprobe analyses; numbers within brackets were obtained from single or a subordinate number of spots

Site	*ZEA*	*ZFA*	*ZFB*	*ZUA*	*ZUB*
Olivine	57	54	55	29	28
Fo (%)	(52)	(59)		(20)	
Augite	64	69	69	65	66
Mg/(Mg + Fe) (mol %)	(65)		(68)	(61)	(58)
Plagioclase	43, 4	37, n.d.	56, 2	37, 5	51, 6
An, Or (%)	(58, 2)		(35–48, 3–5)	(21–48, 5–9)	(17–47, 3–10)

An = anorthite; Fo = forsterite; Or = orthoclase.

titaniferous magnetite in the dolerite samples from sites *ZUA*, *ZUB* and *ZEA* gave TiO_2 contents of 17, 21 and 18 wt.%, respectively, representing the bulk composition of the magnetite–ulvöspinel exsolution. This is in agreement with the microscopically estimated 50%–50% ratio of magnetite and ulvöspinel. In samples from sites *ZFA* and *ZFB* the TiO_2 content of magnetite is approximately 16 wt.%.

3. The remanent magnetization of the Loftahammar gabbro

The alternating magnetic field and thermal demagnetization diagrams of test-specimens from sites in the Loftahammar gabbro are depicted in Fig. 3. Soft-magnetization components in these specimens could be eliminated either by a.f. demagnetization at peak val-

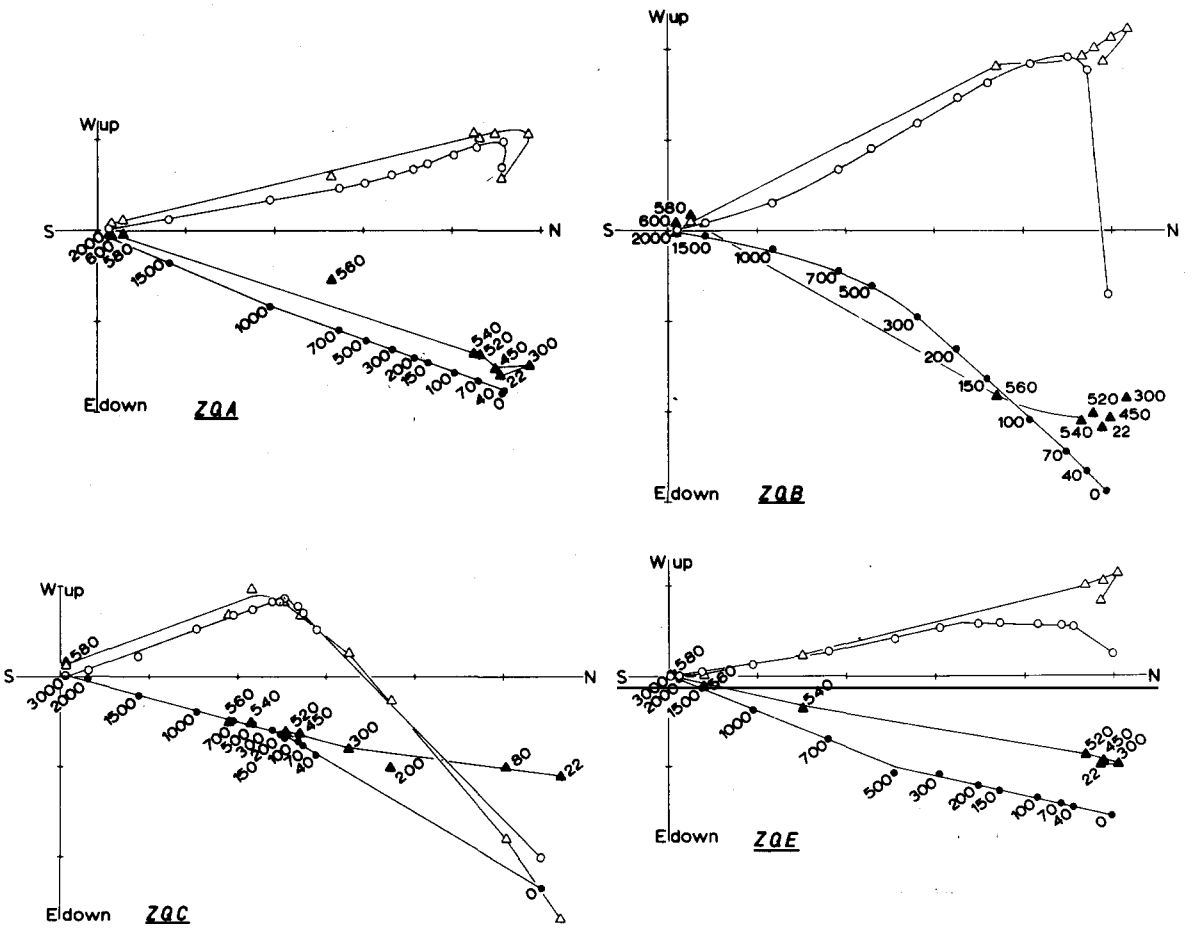


Fig. 3. Orthogonal projections of the end points of the changing resultant natural remanent magnetization vector during progressive alternating magnetic field and thermal demagnetization. Solid symbols plot in the horizontal plane and open symbols plot in the vertical N–S plane. Alternating magnetic field demagnetization diagrams are denoted by circles with peak intensities of the alternating magnetic field in Oersted (Oe); thermal demagnetization diagrams are denoted by triangles with the temperature in °C. The demagnetization diagrams are of test-specimens from samples collected at sites *ZQA*, *ZQB*, *ZQC* and *ZQE* in the Loftahammar gabbro. The diagrams *ZQA*, *ZQB* and *ZQC* are of specimens cut from one sample each, whereas the diagrams *ZQE* are of specimens from different samples. Each unit on either axis represents for the alternating magnetic field and thermal demagnetization diagrams respectively: *ZQA*: $1.0 \cdot 10^{-3}$ and $1.4 \cdot 10^{-3}$ G; ZQB: $0.55 \cdot 10^{-3}$ and $1.2 \cdot 10^{-3}$ G; *ZQC*: $1.0 \cdot 10^{-3}$ and $1.1 \cdot 10^{-3}$ G; *ZQE*: $1.8 \cdot 10^{-3}$ and $1.6 \cdot 10^{-3}$ G.

ues varying between 100 and 200 Oe or by thermal demagnetization up to 300°C. A considerable higher blocking temperature at 520°C for the "soft" component was found in a test-specimen from site *ZQC*. The directions of the "soft", low-coercive–low-temperature components show some scatter, but are generally downward. This suggests a viscous magnetization acquired during recent geological time.

Prolonged demagnetization of test-specimens from the Loftahammar gabbro showed a linear decrease of the remanent magnetization vector, whereas generally better results were obtained by the a.f. demagnetization method than with thermal demagnetization. The direction of the remanent magnetization vector, after elimination of the "soft" component, is similar for test-specimens from all sites in the Loftahammar gabbro, and is therefore called the characteristic remanent magnetization (Zijderveld, 1967). The direction of the characteristic remanent magnetization was obtained by partial a.f. demagnetization of the samples in steps at 200-, 500- and 1,000-Oe peak value, and averaging the directions of the end-point vectors. Samples from site *ZQB* were found to have a compound characteristic magnetization, i.e. a "softer" and a "harder" component with slightly different directions. In this case a "mean characteristic magnetization" was calculated as the average direction of the magnetization vector eliminated between 200 and 1,000 Oe and magnetization vector left after 1,000 Oe. From seven samples collected at site *ZQD*, only two samples were found to have a characteristic remanent magnetization. The remaining samples possessed only a soft magnetization.

The range of blocking temperatures of the characteristic remanent magnetization of the Loftahammar gabbro samples is 540–580°C, thus attaining the Curie point of pure magnetite (580°C). This is in agreement with the mineralogical observations and the low content of TiO_2 in the magnetite of the Loftahammar gabbro.

The mean characteristic magnetization directions and corresponding pole positions are compiled in Table IV. The Rb–Sr total-rock isochron age of $1{,}694 \cdot 10^6$ year for the Loftahammar gneiss adds a maximum age limit to the characteristic remanent magnetization of the Loftahammar gabbro within the gneiss, since the basic bodies are synorogenic-synmetamorphic. The gabbro probably became magnetized during the post-Svecofennian uplift and cooling. There is a K–Ar biotite (and Rb–Sr) age for the rocks within the Västervik area of $1{,}465 \cdot 10^6$ year, which may indicate either the final stage of the post-metamorphic cooling or a separate metamorphic Gothian event (according to Priem et al., 1969). In both cases this age forms a

TABLE IV

Mean characteristic magnetization directions and corresponding pole positions of the Loftahammar gabbro

Site	(*n*)	Intensity of NRM (G)	*Q* value	Site mean direction of the characteristic magnetization				Pole position (°)			
				decl. (°)	incl. (°)	confidence α_{95} or A_{95} (°)	precision *k*	lat.	long.	d.p.	d.m.
ZQA	7	10^{-3}–10^{-2}	0.5–3.6	20.9	−9.9	3	366	25N	174E	2	3
ZQB *1	5	10^{-3}–10^{-2}	2.3–9.0	18.8	−16.0	4	322	22N	176E	2	4
ZQB *2				22.8	−17.2	9	80	–	–	–	–
ZQB *3				14.7	−14.9	4	300	–	–	–	–
ZQC	6	10^{-3}–10^{-2}	0.6–2.4	13.2	−17.4	2	1,832	22N	178W	1	2
ZQD	2	10^{-5}–10^{-3}	0.3–2.0	15.0	−19.0	–	–	21N	180E	–	–
ZQE	6	10^{-3}–10^{-2}	6.0–19	13.3	−13.0	5	208	25N	178W	2	5
Average sites (N = 5)				16.3	−15.0	4.6	280	23N	179E	2.4	4.7
Average samples (n = 26)				16.5	−14.2	2.1	181	22N	178E	1.1	2.2

*1 Average magnetization direction elicitated between a.f. demagnetization at 200 and 1,000 Oe and after 1,000 Oe.
*2 Magnetization direction between 200 and 1,000 Oe.
*3 Magnetization direction after 1,000 Oe.

lower limit of the magnetization age of the characteristic remanence found in the Loftahammer gabbro, since there is no evidence for a later thermal event of any significance in this area.

4. The remanent magnetization of the Jotnian dolerites

The alternating magnetic field and thermal demagnetization diagrams for test-specimens from the Jotnian dolerites are shown in Fig. 4. Diagram *ZFA* is representative for the type of magnetization found in specimens from both sites *ZFA* and *ZFB*, and diagram *ZUA* for both site *ZUA* and *ZUB*.

The diagrams show soft downwards-directed components, probably of Recent origin, which could be eliminated completely either by a.f. demagnetization at peak values varying between 100 and 200 Oe or by thermal demagnetization up to 400 or 500°C. The high-coercive–high-temperature magnetization components have similar directions for the various sites within the Jotnian dolerites and are therefore considered to be characteristic. As samples of the Loftahammar gabbro, a.f. demagnetization yielded better results than thermal demagnetization.

The blocking temperatures of the characteristic remanent magnetization in test-specimens from the Jotnian dolerites are lower than the Curie point of pure magnetite. The demagnetization diagrams *ZFA* anol *ZUA* (Fig. 4) show that the larger part of the characteristic magnetization was eliminated between 450 and 540°C; for diagram *ZEA* (Fig. 4) these temperatures were 500 and 560°C. It is known that both the TiO_2 content of the magnetic phase and the oxidation state are effective in controlling the blocking temperatures (e.g. Stacey and Banerjee, 1974). Since the blocking temperatures determined in the test-specimens from sites *ZFA* and *ZFB* are higher than would be expected on account of the observed TiO_2 content in a homogeneous magnetite phase, the tentative conclusion is that the titaniferous magnetite in these rocks consists of a sub-microscopic magnetite–ulvöspinel exsolution. The observed lower

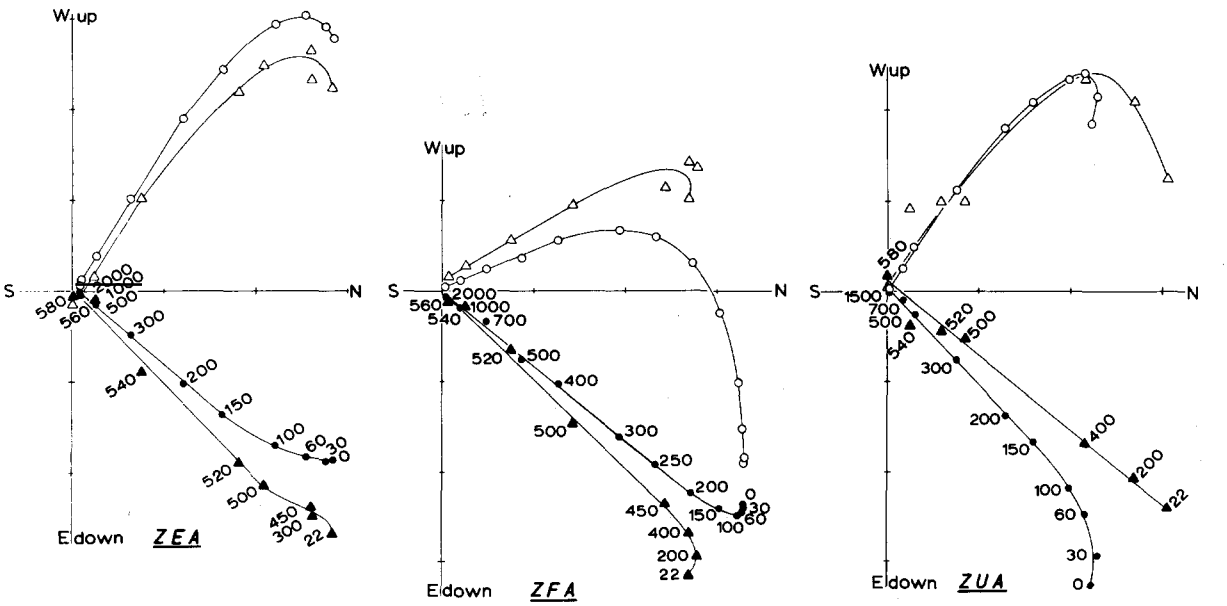


Fig. 4. Demagnetization diagrams of test-specimens from samples collected at sites *ZEA*, *ZFA* and *ZUA* in Jotnian dolerites. The diagrams *ZEA* and *ZUA* are of specimens cut from one sample each, whereas the diagrams *ZFA* are of specimens from different samples. Mode of projection as in Fig. 3. Each unit on either axis represents for the alternating magnetic field and thermal demagnetization diagrams, respectively: *ZEA:* $3.7 \cdot 10^{-4}$ and $3.8 \cdot 10^{-4}$ G; *ZFA*: $4.4 \cdot 10^{-4}$ and $3.7 \cdot 10^{-4}$ G; *ZUA*: $4.2 \cdot 10^{-4}$ and $4.0 \cdot 10^{-4}$ G.

blocking temperatures as compared with the blocking temperatures found for pure magnetite (e.g. the magnetite with 0.1 wt.% or less TiO_2 in the Loftahammar gabbro) can be explained by assuming about 10% of the ulvöspinel molecules dissolved in the magnetite phase, according to the diagram given by O'Reilly and Readman (1971).

The direction of the characteristic remanent magnetization component was determined for each sample by subtraction of the end-point vectors which determine the linear part of the characteristic magnetization. For example, in case of the test-specimens from site *ZEA* (Fig. 4) for the direction of the characteristic magnetization the direction is taken of the difference vector between the two end-point vectors obtained after a.f. demagnetization at 200 and 1,000 Oe, respectively. This method was followed since in a number of cases the line connecting the successive end points of the decreasing magnetization vector did not dissect the origin of the cartesian coordinates (used in the representation of the demagnetization diagrams). This is probably caused by excentricity of the magnetization (Zijderveld, 1975) due to inhomogeneous partition of magnetite concentrations in the specimens.

The mean characteristic magnetization directions and corresponding pole positions for the Jotnian dolerites are listed in Table V. Sites *ZUA* and *ZUB* are relative close within the Nordingrå dolerite and the results are therefore averaged. Sites *ZFA* and *ZFB* are probably in closely related dolerite bodies in the Gävle area, and the results from these sites are also averaged. The characteristic magnetization directions from these sites have lower inclinations than were found in the dolerite at Gnarp and Nordingrå. Some time might have elapsed between the intrusion of the Nordingrå dolerite and the related Gnarp dolerites, and the intrusion of the dolerite at Gävle, and this is one possible explanation for the different inclinations.

There is no mineralogical evidence that the Jotnian dolerites presently studied have been overprinted by a metamorphic event. According to Welin and Lundqvist (1975), the K–Ar isochron age of $1{,}245 \cdot 10^6$ year is regarded as the time of retention of Ar and most probably corresponds to the crystallization of the Nordingrå dolerite. We consider the age of $1{,}245 \cdot 10^6$ year therefore as the best estimate for the age of the characteristic remanent magnetization of the Nordingrå (and Gnarp) dolerite.

5. Palaeomagnetic correlation

Palaeomagnetic pole positions of Jotnian sediments and dolerites from Sweden and Finland are compiled in Table VI. It shows the good agreement of pole positions from Jotnian dolerites on both sides of the Bothnian Sea (Fig. 1). These include the dolerites at Nordingrå, Gnarp and Gävle (Sweden) and at Vaasa,

TABLE V

Mean characteristic magnetization directions and corresponding pole positions of the Jotnian dolerites

Site	(*n*)	Intensity of NRM (G)	*Q* value	Site mean direction of the characteristic magnetization				Pole position (°)			
				decl. (°)	incl. (°)	confidence α_{95} (°)	precision *k*	lat.	long	d.p.	d.m.
ZEA	6	10^{-3}	1.2–2.0	45.8	−51.8	8	78	11S	159E	7	10
ZFA	6	10^{-3}	1.9–2.6	46.3	−17.3	4	282	12N	150E	2	4
ZFB	9	10^{-3}	1.1–1.8	50.1	−29.0	3	221	4N	149E	2	4
ZUA	6	10^{-3}	0.6–1.0	45.8	−52.5	5	212	13S	160E	4	6
ZUB	6	10^{-3}–10^{-2}	0.7–1.1	45.4	−32.4	9	56	2N	156E	6	10
Average *ZFA* and *ZFB* (sites)				48.1	−23.3	–	–	8N	150E	–	–
Average *ZFA* and *ZFB* (samples)				48.5	−24.3	4	104	7N	150E	2	4
Average *ZUA* and *ZUB* (sites)				45.6	−42.4	–	–	5S	158E	–	–
Average *ZUA* and *ZUB* (samples)				45.6	−42.5	7	37	5S	158E	5	9

TABLE VI

Palaeomagnetic pole positions of Jotnian rocks from Sweden and Finland

Name	Sites *N*	Samples *n*	Pole positions				References
			lat. (°)	long. (°)	d.p. (°)	d.m. (°)	
Satakunta sandstone, Finland	6	10	3N	180E	8	12	Neuvonen (1973)
Dalarna basalts, Sweden	2	11	32N	174W	–	–	Mulder (1971)
Dalarna dolerites, Sweden	4	19	23N	178E	21	40	Mulder (1971)
Vaasa dolerites, Finland (*VD*)	14	15	7N	164E	3	6	Neuvonen (1966)
Satakunta dolerites, Finland (*SD*)	15	18	2N	158E	3	5	Neuvonen (1965)
Märket dolerites, Finland (*MD*)	7	9	6S	146E	7	11	Neuvonen and Grundström (1969)
Gnarp dolerite, Sweden (*ZE*)	1	6	11S	159E	7	10	this study
Gävle dolerite, Sweden (*ZF*)	2	15	7N	150E	2	4	this study
Nordingrå dolerite, Sweden (*ZU*)	2	12	5S	158E	5	9	this study
Average pole position of *VD, SD, MD, ZE, ZF, ZU (D)*			1S	156E	$A_{95} = 8°$ ($N = 6$)		

Satakunta and Märket (Finland). Apparently, these dolerites are approximately contemporaneous. The overall pole position of the Jotnian dolerites from around the Bothnian Sea is situated at 1°S, 156°E (Table VI), whereas the age assigned to this pole follows from the radiometric evidence as being (1,300 –1,200) · 10^6 years.

The pole position from the Satakunta sandstone in Finland is situated not far from the pole position observed for the Jotnian dolerites around the Bothnian Sea. This is in agreement with the ranges of radiometric ages found for the Jotnian sediments and dolerites (Lundqvist, 1968), indicating that not much time elapsed between deposition of the sediments and intrusion of the dolerites.

Mulder (1971) investigated Jotnian basalts and dolerites as well as sub-Jotnian Upper-Dala volcanics (Table VII) in the Dalarna district of Sweden. He found pole positions for these rocks situated not far from pole positions of Palaeozoic rocks in Scandinavia. Because of the proximity of the Caledonian front, Mulder concluded that the remanent magnetization of the rocks in the Dalarna district may have been overprinted during the Caledonian period.

On the other hand the pole position of the Upper-Dala volcanics and the Loftahammar gabbro (Table VII) are almost identical, which, taking into account the similar Rb–Sr ages, can hardly be regarded as a mere coincidence, since the Loftahammar gabbro is not likely to have been overprinted by the Caledonian metamorphism. Therefore, the similarity of pole positions favours a pre-Caledonian age for the characteristic magnetization of the Upper-Dala volcanics as well. Consequently, the pole positions of the basalts and dolerites from the Dalarna district as found by Mulder (1971), could be considered to have a pre-Caledonian age too. In that case the palaeomagnetic evidence, in view of the apparent polar wandering path according to Neuvonen (1970) (Fig. 2), suggests that the characteristic remanent magnetization of the Jotnian basalts and some of the dolerites (Särna type) of the Dalarna district are older than those of the Jotnian dolerites around the Bothnian Sea. One of the dolerites investigated by Mulder (1971) in the Dalarna district (Åsby type), however, has a characteristic magnetization direction similar as that found in the Jotnian dolerites of the present study and those from Finland. This suggests that the Åsby dolerite is younger than the Särna dolerites and the Öje basalts in the Dalarna district.

6. Pole positions from the Baltic Shield for the (1,900–1,200) · 10^6-year interval and a tentative reconstruction of the North Atlantic area

Palaeomagnetic pole positions from the Baltic Shield ranging in age from approximately 1,900 · 10^6 to

TABLE VII

Palaeomagnetic pole positions of Svecofennian and Jotnian rocks from Finland and Sweden

Symbol Fig. 5	Name	Age (10^6 years)	Sites *N*	Samples *n*	Pole positions (°)				References
					lat.	long.	d.p.	d.m.	
GD	gabbro-diorite intrusions, Finland and Sweden	1,900	7		45N	119W	$A_{95} = 7°$		Neuvonen (1974)
AV	Åva intrusive, Finland	1,830	15	15?	41N	169E	9	17	Neuvonen (1970)
K	Kumlinge dolerites, Finland	1,650 minimum	7	7?	13N	159W	11	19	Neuvonen (1970)
F	Föglödolerite, Finland	1,650 minimum	8	8	31N	174W	6	12	Neuvonen and Grundström (1969)
DV	Upper-Dala volcanics, Sweden	1,669 maximum	2	11	23N	176E			Mulder (1971); Priem et al. (1970)
LG	Loftahammar gabbro, Sweden	1,694 maximum 1,460 minimum	5	26	23N	179E	2	5	this study, Table IV
DB	Dalarna basalts, Sweden	"Jotnian"	2	11	32N	174W	–	–	Mulder (1971)
DD	Dalarna dolerites, Sweden	"late-Jotnian"	4	19	23N	178W	21	40	Mulder (1971)
SS	Satakunta sandstone, Finland	1,470–1,420	6	10	3N	180E	8	12	Neuvonen (1974)
D	dolerites, Finland and Sweden	1,300–1,200	6	75	1S	156E	$A_{95} = 8°$		this study, Table VI

1,200 · 10^6 year are summarized in Table VII and plotted in Fig. 5, together with palaeomagnetic pole positions from the North American Shield, covering the same time interval.

The beginning and end of the apparent polar track of the Baltic Shield for this time interval are marked by two "master" poles: the overall palaeomagnetic pole position for gabbro-diorite intrusions in Sweden and Finland, pole *GD*, with an estimated age of 1,900 · 10^6 year (Neuvonen, 1974), and the overall palaeomagnetic pole position for the Jotnian dolerites around the Bothnian Sea, in Sweden and Finland, pole *D*, with an estimated age of (1,300–1,200) · 10^6 year (this study, including results from Neuvonen (1970)). The other palaeomagnetic poles from the Baltic Shield with intermediate ages, appear to be situated between the two "master" poles. Thus, the apparent polar wander curve suggested by Neuvonen (1970), shown in Fig. 2, is a correct indication of the trend of apparent polar wander from 1,900 · 10^6 to 1,200 · 10^6 year relative to the Baltic Shield, although the history of apparent polar movement during the intervening period is largely unknown.

It is tempting to use the palaeomagnetic data now available from the Baltic Shield for a reconstruction of its relative position to the North American Shield at the time of the Svecofennian orogeny and the following anorogenic Jotnian period, which can be correlated geologically and radiometrically with the Hudsonian orogeny and the Mackenzie period in North America. Fig. 6 shows the rearrangement of pole positions from the Baltic Shield after rotation in order to match these poles with those from the North American Shield. The reconstructed position of the Baltic Shield according to this operation is illustrated in Fig. 7. In this picture the position of Greenland is corrected for Palaeozoic and post-Palaeozoic drift according to the so-called Bullard fit (Bullard et al., 1965). Essential in this re-

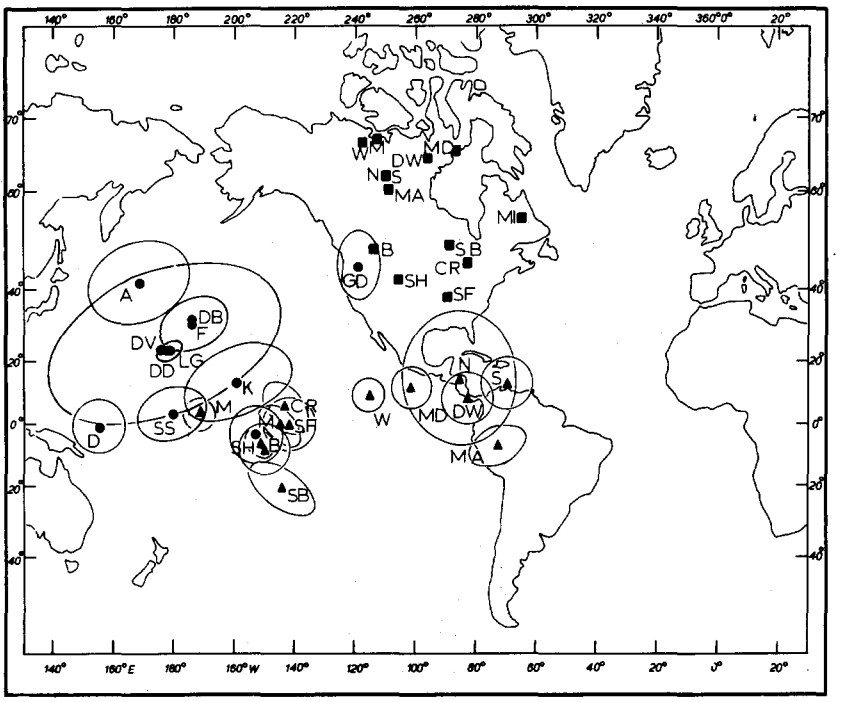


Fig. 5. Pole positions between 1,900 · 10^6 and 1,200 · 10^6 year from the Baltic Shield (dots) and the North American Shield (trianlges). Sampling locations for the first shown in Fig. 1; sampling locations for the latter are indicated by squares. Indices and ellipses or circles of confidence for the poles from the Baltic Shield are indicated according to Table VII. Data for the North American poles are from the Ottawa catalogue of palaeomagnetic directions and poles (first issue from Hicken et al. (1972); second issue from Irving and Hastie (1975)). Indices and catalogue numbers: *S*, 1-410; *DW*, 1-481; *N*, 1-411; *MA*, 1-468; *MD*, 1-456; *W*, 1-504; *CR*, 1-129; *MI*, 1-205; *SF*, 1-240; *SH*, 1-172; *B*, 1-225; *SB*, 1-473; *M*, 1-340.

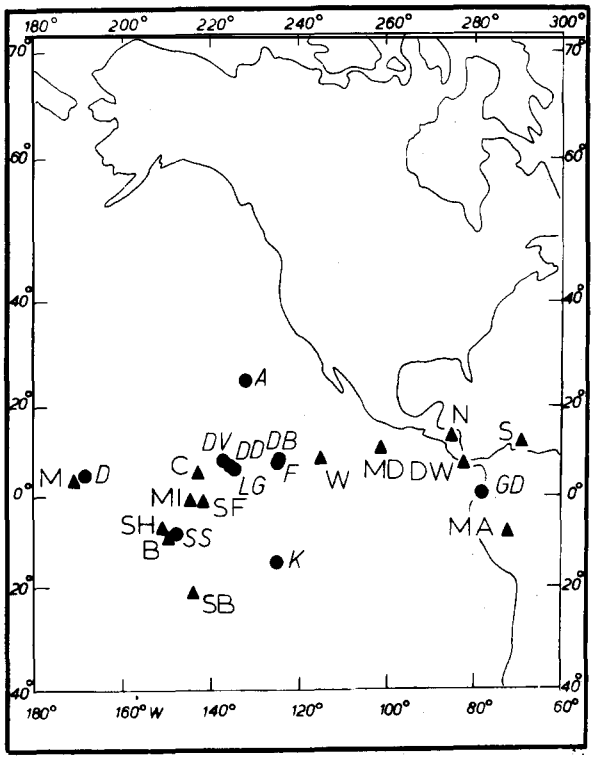


Fig. 6. Matching of pole positions from the North American and Baltic shields for the (1,900–1,200) · 10^6-year interval. Indices as in Fig. 5.

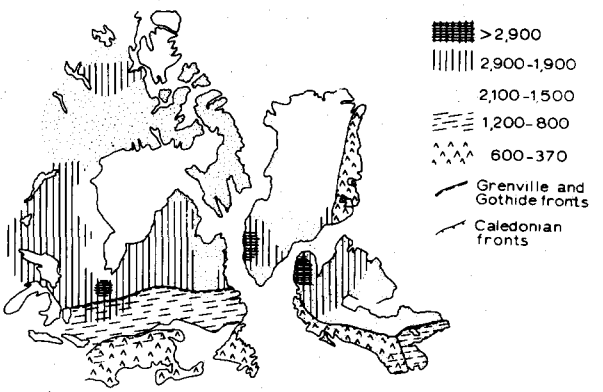


Fig. 7. Reconstruction of the position of the Baltic Shield with respect to the North American Shield for the (1,900 –1,200) · 10^6-year interval according to the matching of pole positions shown in Fig. 6. The position of Greenland is corrected for Palaeozoic and Mesozoic drift (Bullard et al., 1965).

construction is the contiguity of the pre-Sveconorwegian, pre-Grenville orogenic belts, suggesting that the Precambrian shields of the Baltic and of North America and Greenland formed a crustal unit from the Early Precambrian until approximately 1,200 · 10^6 year.

The crustal unity of the Baltic and the North American and Greenland shields probably came to an end after or even during the last stage of the Jotnian period. This follows from the opposite directed "hair-pin"-like changes (Irving and Park, 1972) of the apparent polar wander tracks for the North American Shield (e.g., Donaldson et al., 1973) and the Baltic Shield (Fig. 2) at approximately 1,200 · 10^6 year ago. The breaking-up of the crustal unity may have been accompanied with the widespread occurrence of basic intrusions between 1,300 · 10^6 and 1,200 · 10^6 year ago, like the Jotnian dolerites in Sweden and Finland, the Mackenzie intrusives in Canada and the Elsonian intrusions in south Greenland and in Labrador.

The changing trend of the apparent polar wander relative to the Baltic Shield and the North American Shield at approximately 1,200 · 10^6 year ago could be considered indicative for the tectonic reorganization preceding and during the cataclysm of the Sveco-norwegian–Grenville orogeny in the North Atlantic area. Possibly the Central Schistosity Zone of southern Sweden (Magnusson, 1965) started to be created in that time.

Some tentative reconstructions of the new configuration of the Baltic Shield and the North American and Greenland shields after the Sveconorwegian–Grenville orogeny for the (1,000–600) · 10^6 -year interval have been given by Ueno et al. (1975). A comparison of the reconstruction of the Baltic Shield for the (1,900 –1,200) · 10^6-year interval, suggested in this study, and the reconstructions as suggested for the (1,000 –600) · 10^6-year interval by Ueno et al. (1975) shows a clockwise rotation over approximately 90° taking place from 1,200 · 10^6 until 1,000 · 10^6 year ago. This conclusion, if based on largely correct reconstructions, can be helpfull in studying the tectonic style of the Sveconorwegian and Grenville orogens.

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